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Assignment - 17.1

Batch - 14

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Task 1 – Customer Data Cleaning

Prompt:

Generate a Python script to clean a customer dataset.
Handle missing values in age, income, city columns, remove duplicates, standardize city names to lowercase, and encode categorical variables gender and city.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
data = pd.read_csv("/content/shopping_trends.csv")
data['Age'].fillna(data['Age'].median(), inplace=True)
data.drop_duplicates(inplace=True)
data['Location'] = data['Location'].str.lower()
le = LabelEncoder()
data['Gender'] = le.fit_transform(data['Gender'])
data['Location'] = le.fit_transform(data['Location'])
print(data.head())
```

Output:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	\
0	1	55	1	Blouse	Clothing	53	
1	2	19	1	Sweater	Clothing	64	
2	3	50	1	Jeans	Clothing	73	
3	4	21	1	Sandals	Footwear	90	
4	5	45	1	Blouse	Clothing	49	

	Location	Size	Color	Season	Review Rating	Subscription Status	\
0	16	L	Gray	Winter	3.1	Yes	
1	18	L	Maroon	Winter	3.1	Yes	
2	20	S	Maroon	Spring	3.1	Yes	
3	38	M	Maroon	Spring	3.5	Yes	
4	36	M	Turquoise	Spring	2.7	Yes	

	Payment Method	Shipping Type	Discount Applied	Promo Code Used	\
0	Credit Card	Express	Yes	Yes	
1	Bank Transfer	Express	Yes	Yes	
2	Cash	Free Shipping	Yes	Yes	
3	PayPal	Next Day Air	Yes	Yes	
4	Cash	Free Shipping	Yes	Yes	

	Previous Purchases	Preferred Payment Method	Frequency of Purchases
0	14	Venmo	Fortnightly
1	2	Cash	Fortnightly
2	23	Credit Card	Weekly
3	49	PayPal	Weekly
4	31	PayPal	Annually

Explanation:

- Missing values replaced using median/mode
- Duplicate rows removed
- City names converted to lowercase
- Gender and city converted to numeric form for ML models

Task 2 – Sales Data Preprocessing

Prompt:

Generate a Python script to preprocess sales data.
Convert date column to datetime, create month/year/day features,
normalize sales amount, and remove outliers.

Code:

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
data = pd.read_csv("/content/sales_data.csv")
data.columns = data.columns.str.strip()
print("Columns in dataset:")
print(data.columns)
date_col = None
for col in data.columns:
    if "date" in col.lower():
        date_col = col
        break
if date_col is None:
    print("No date column found in dataset")
else:
    print("Date column found:", date_col)
    data[date_col] = pd.to_datetime(data[date_col])
    data['Month'] = data[date_col].dt.month
    data['Year'] = data[date_col].dt.year
    data['DayOfWeek'] = data[date_col].dt.dayofweek
sales_col = None
for col in data.columns:
    if "sale" in col.lower() or "revenue" in col.lower():
        sales_col = col
        break
if sales_col is not None:
    scaler = MinMaxScaler()
    data[sales_col] = scaler.fit_transform(data[[sales_col]])
    print("Normalized:", sales_col)
else:
    print("No sales column found")
print(data.head())
```

Output:

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	\
0	1052	0.090411	Bob	North	5053.97	18	
1	1093	0.301370	Bob	West	4384.02	17	
2	1015	0.720548	David	South	4631.23	30	
3	1072	0.643836	Bob	South	2167.94	39	
4	1061	0.224658	Charlie	East	3750.20	13	

	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	\
0	Furniture	152.75	267.22	Returning	0.09	
1	Furniture	3816.39	4209.44	Returning	0.11	
2	Food	261.56	371.40	Returning	0.20	
3	Clothing	4330.03	4467.75	New	0.02	
4	Electronics	637.37	692.71	New	0.08	

	Payment_Method	Sales_Channel	Region_and_Sales_Rep	Month	Year	DayOfWeek
0	Cash	Online	North-Bob	2	2023	4
1	Cash	Retail	West-Bob	4	2023	4
2	Bank Transfer	Retail	South-David	9	2023	3
3	Credit Card	Retail	South-Bob	8	2023	3
4	Credit Card	Online	East-Charlie	3	2023	4

Explanation:

- Datetime conversion enables time analysis
- Feature engineering improves model performance
- Normalization scales data uniformly
- IQR method removes extreme values

Task 3 – Healthcare Data Preparation

Prompt:

Generate a Python script to preprocess healthcare data by handling missing values, encoding categorical features, scaling numerical data, and splitting into training and testing sets.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
data = pd.read_csv("/content/dirty_v3_path.csv")
data.columns = data.columns.str.strip()
print("Columns in dataset:")
print(data.columns)
num_cols = data.select_dtypes(include=['int64', 'float64']).columns
for col in num_cols:
    data[col].fillna(data[col].mean(), inplace=True)
cat_cols = data.select_dtypes(include='object').columns
for col in cat_cols:
    data[col].fillna(data[col].mode()[0], inplace=True)
le = LabelEncoder()
for col in cat_cols:
    data[col] = le.fit_transform(data[col])
scaler = StandardScaler()
data[num_cols] = scaler.fit_transform(data[num_cols])
X = data.iloc[:, :-1]
y = data.iloc[:, -1]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
print("Training shape:", X_train.shape)
```

Output:

```
Columns in dataset:
Index(['Age', 'Gender', 'Medical Condition', 'Glucose', 'Blood Pressure',
       'BMI', 'Oxygen Saturation', 'LengthOfStay', 'Cholesterol',
       'Triglycerides', 'HbA1c', 'Smoking', 'Alcohol', 'Physical Activity',
       'Diet Score', 'Family History', 'Stress Level', 'Sleep Hours',
       'random_notes', 'noise_col'],
      dtype='object')
/tmp/ipykernel_17029/1863719885.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy. For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead.

    data[col].fillna(data[col].mean(), inplace=True)
/tmp/ipykernel_17029/1863719885.py:13: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy. For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead.

    data[col].fillna(data[col].mode()[0], inplace=True)
Training shape: (22340, 19)
```

Explanation:

- Missing medical values replaced using mean
- Categorical variables converted to numeric
- Standard scaling improves ML accuracy
- Train-test split prevents overfitting

TASK 4 – Real-Time Application: Multi-Language Snippet Converter

Prompt:

Generate a Python script to preprocess employee salary data by handling missing values, removing salary outliers, standardizing department names, and encoding location and department.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
data = pd.read_csv("/content/Employee_Salary.csv")
data.fillna(method='ffill', inplace=True)
Q1 = data['Salary'].quantile(0.25)
Q3 = data['Salary'].quantile(0.75)
IQR = Q3 - Q1
data = data[(data['Salary'] >= Q1 - 1.5 * IQR) &
            (data['Salary'] <= Q3 + 1.5 * IQR)]
data['Department'] = data['Department'].str.lower()
le = LabelEncoder()
data['Location'] = le.fit_transform(data['Location'])
data['Department'] = le.fit_transform(data['Department'])
print(data.head())
```

Output:

	Employee_ID	Department	Job_Title	Experience	Education	Location	Salary
0	101	1	Manager	10	MBA	2	75000
1	102	2	Developer	5	B.Tech	4	60000
2	103	0	Analyst	7	MBA	1	65000
3	104	2	Developer	3	B.Tech	0	55000
4	105	1	Executive	4	BBA	2	50000

Explanation:

- Missing values filled using forward fill
- IQR removes unrealistic salary values
- Lowercase standardizes department names
- Encoding converts text to numbers for analytics